Chapitre 8 : Fonctions

Solution des exercices

Exercice 1

```
def surfaceRectangle(longeur, largeur):
    return longeur * largeur

print(surfaceRectangle(2,4))
```

Exercice 2

```
def ligneCar(n, ca):
    return ca * n

print(ligneCar(3, "Allo"))
```

Exercice 3

```
from math import pi

def surfCercle(R):
    return pi * R**2

print(surfCercle(2.5))
```

Exercice 4

```
def volBoite(x1,x2,x3):
    return x1*x2*x3

print(volBoite(5.2, 7.7, 3.3))
```

Exercice 5

```
def maximum(n1,n2,n3):
    if n1 > n2 and n1 > n3:
        return n1
    if n2 > n1 and n2 > n3:
        return n2
    else:
        return n3

print(maximum(2, 5, 4))
```

Exercice 6

```
def indiceMax(liste):
    max = liste[0]
    for entier in liste:
        if max < entier:
            max = entier

    return liste.index(max)

serie = [5, 8, 2, 1, 9, 3, 6, 7]
print(indiceMax(serie))
```

Exercice 7

```
def inverse(ch):
    chInv = ""
    for car in ch:
        chInv = car + chInv
    return chInv


print(inverse("allo"))
```

Exercice 8

```
def compteMots(ph):
    tabMot = ph.split()
    return len(tabMot)

print(compteMots("allo allo"))
```

Exercice 9

```
def nomMois(n):
    mois = ["Janvier", "Fevrier", "Mars", "Avril", "Mai", "Juin"]
    return mois[n-1]

print(nomMois(4))
```

Exercice 10

```
def nomJours(n):
    jours = ["Lundi", "Mardi", "Mercredi", "Jeudi", "Vendredi", "Samedi",
"Dimanche"]
    return jours[n]

print(nomJours(0))
```

Exercice 11

```
from datetime import date

def premierJours(annee, mois):
    maDate = date(annee, mois, 1)
    return maDate.weekday()

print(premierJours(2014, 1))
```

Exercice 12

```
def nbJours(annee, mois):
    if mois == 2:
        if (((annee % 4) == 0) and ((annee % 100 != 0) or (annee % 400 == 0))):
            return 29
        return 28
    elif (mois == 1 or mois == 3 or mois == 5 or mois == 7 or mois == 8 or mois
== 10 or mois==12):
        return 31
    else:
        return 30

print(nbJours(2014, 1))
```

Exercice 13

```
from datetime import date

def premierJours(annee, mois):
    maDate = date(annee, mois, 1)
    return maDate.weekday()

def nbJours(annee, mois):
    if mois == 2:
        if (((annee % 4) == 0) and ((annee % 100 != 0) or (annee % 400 ==
0))):
            return 29
        return 28
    elif (mois == 1 or mois == 3 or mois == 5 or mois == 7 or mois == 8 or
mois == 10 or mois==12):
        return 31
    else:
        return 30

def calendrier(annee, mois):
    cal = "lun\tmar\tmer\tjeu\tven\tsam\tdim\n"
    for num in range(0, premierJours(annee, mois) ):
        cal = cal + " \t"
    for num in range(1, nbJours(annee, mois)+1):
        cal = cal + str(num)
        if (num + premierJours(annee, mois))% 7 == 0:
            cal = cal + "\n"
        else:
            cal = cal + "\t"
```

```
    return cal

print(calendrier(2014, 1))
```

Exercice 13

```
from datetime import date

def premierJours(annee, mois):
    maDate = date(annee, mois, 01)
    return maDate.weekday()

def nbJours(annee, mois):
    if mois == 2:
        if (((annee % 4) == 0) and ((annee % 100 != 0) or (annee % 400 == 0))):
            return 29
        return 28
    elif (mois == 1 or mois == 3 or mois == 5 or mois == 7 or mois == 8 or mois
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def calendrier(annee, mois):
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    for num in range(0, premierJours(annee, mois) ):
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    for num in range(1, nbJours(annee, mois)+1):
        cal = cal + str(num)
        if (num + premierJours(annee, mois))% 7 == 0:
            cal = cal + "\n"
        else:
            cal = cal + "\t"

    return cal

print calendrier(2014, 1)
```